

Where irrigation exists is globally contested

5. Analyses supporting the reply to the reviewers

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1 Preliminary

```
# ANALYSES SUPPORTING THE REPLY TO THE REVIEWERS #####
#####

# This script reproduces every quantitative figure cited in our reply to the
# reviewers. All analyses use the harmonized ensemble at 0.2 deg restricted to
# cropland cells (the same universe as the main analysis).

# Load libraries -----

sensobol::load_packages(c("data.table", "here", "terra", "magrittr"))

# Functions reused from the main analysis -----

source(here("functions", "filter_long_dt_by_crop_mask_aggregated.R"))
source(here("functions", "compute_tau_max_disagreement_fun.R"))

# Harmonized ensemble at 0.2 deg, restricted to cropland -----

dt <- fread(here("datasets", "irrigated_areas_regridded",
                 "irrigated_areas_regridded_02.csv"))
dt[, resolution := "0.2deg"]
crop <- rast(here("irrigated_area_datasets", "GirrEO", "asap_mask_crop_v02.tif"))
dt <- filter_long_dt_by_crop_mask_aggregated(dt = dt, crop_native = crop,
                                             rule = "any")

# Cells x maps -----

wide <- dcast(dt, lon + lat + country ~ dataset, value.var = "mha", fill = 0)

maps <- c("gaez_v4", "giam", "gmia", "gripc", "luh2", "meier",
          "mirca2000", "mirca_os", "nagaraj", "spam")
stopifnot(all(maps %in% names(wide)))

# Nominal reference year of each map (Table on temporal characteristics) -----

nominal <- c(giam = 2000, mirca2000 = 2000, gmia = 2005, gripc = 2005,
             nagaraj = 2005, meier = 2005, spam = 2010, mirca_os = 2015,
             gaez_v4 = 2015, luh2 = 2015)

# Mapping families -----

census <- c("gmia", "mirca2000", "mirca_os", "spam", "gaez_v4") # FAO-statistics
rs      <- c("giam", "gripc") # remote sensing

# Detectability threshold grid (1 ha to 1e5 ha) and 1%-of-cell tolerance -----
```

```

taus_ha      <- c(0, 10^(seq(0, 5, by = 0.1)))
taus_mha     <- taus_ha * 1e-6
zero_tol_02  <- 500 # 1% of the 0.2 deg cell, in ha

# Existential disagreement: of the cells where at least one map records
# irrigation (> 0), the fraction in which at least one map records exactly zero.

exist_union <- function(w, cols) {
  A <- as.matrix(w[, ..cols])
  n_present <- rowSums(A > 0, na.rm = TRUE)
  disagree <- n_present > 0 & n_present < length(cols)
  Amin <- do.call(pmin, c(as.data.frame(A), list(na.rm = TRUE)))
  sum(disagree & Amin == 0) / sum(n_present > 0)
}

# Extreme disagreement: fraction of cells whose tau_max exceeds ten times the
# detectability threshold (one map reports > ~10% of the cell, another ~0).

extreme_frac <- function(w, cols, zero_tol = zero_tol_02) {
  tm <- compute_tau_max_existential_fun(w, cols, taus_mha, taus_ha,
                                         zero_tol_ha = zero_tol)
  mean(!is.na(tm$tau_max_ha) & tm$tau_max_ha > 10 * zero_tol)
}

```

2.1 R1.1: the detectability threshold as a fraction of the grid cell

```
# Equator cell area (ha) per resolution -----
R_earth <- 6371
deg <- c("0.2" = 0.2, "0.4" = 0.4, "1" = 1)
cell_ha <- sapply(deg, function(d) (d * pi / 180 * R_earth)^2 * 100)
round(cell_ha)
```

```
# Key tau values as a percentage of the cell -----
tau_key <- c(100, 500, 1000, 2500, 5000, 10000, 25000)
tau_tab <- data.table(
  `tau (ha)` = tau_key,
  `0.2 deg (%)` = round(100 * tau_key / cell_ha["0.2"], 2),
  `0.4 deg (%)` = round(100 * tau_key / cell_ha["0.4"], 2),
  `1 deg (%)` = round(100 * tau_key / cell_ha["1"], 2))

knitr::kable(tau_tab, caption = "Detectability threshold tau as a percentage of th
```

τ (ha)	0.2 deg (%)	0.4 deg (%)	1 deg (%)
100	0.20	0.05	0.01
500	1.01	0.25	0.04
1000	2.02	0.51	0.08
2500	5.05	1.26	0.20
5000	10.11	2.53	0.40
10000	20.22	5.05	0.81
25000	50.55	12.64	2.02

We test whether the disagreement reflects the different moments the maps represent. Existence of irrigation is the metric throughout (see Preliminary).

4

```

set = c("Same year (2005)", "Full ensemble"),
maps = c(length(sameyear_2005), length(maps)),
existential_pct = round(100 * c(exist_union(wide, sameyear_2005),
                                exist_union(wide, maps))))

##           set maps existential_pct
##           <char> <int>           <num>
## 1: Same year (2005)      4           69
## 2:   Full ensemble     10           90

# Recompute over every four-map combination to hold ensemble size fixed -----

combs4 <- combn(maps, 4, simplify = FALSE)
ev4    <- sapply(combs4, function(cc) exist_union(wide, cc))
span4  <- sapply(combs4, function(cc) diff(range(nominal[cc])))
same_ev <- exist_union(wide, sameyear_2005)

data.table(
  quantity = c("Minimum", "Median", "Maximum",
               "Same-year (2005) combination", "Most dispersed (1999-2015)"),
  existential_pct = round(100 * c(min(ev4), median(ev4), max(ev4),
                                same_ev, mean(ev4[span4 == max(span4)]))),
  percentile = c(NA, NA, NA, round(100 * mean(ev4 <= same_ev)), NA))

##           quantity existential_pct percentile
##           <char>           <num>           <num>
## 1:           Minimum             15           NA
## 2:           Median             74           NA
## 3:           Maximum             89           NA
## 4: Same-year (2005) combination      69           36
## 5:   Most dispersed (1999-2015)      74           NA

# Direction of disagreement: under real expansion, a recent (2015) map should
# not record zero where an older (2000-2005) map detects irrigation.

old_t    <- c("giam", "mirca2000", "gmia", "gripc") # 2000-2005
recent_t <- c("gaez_v4", "luh2", "mirca_os")        # 2015

pooled_blind <- function(new) {
  A_old <- as.matrix(wide[, ..old_t])
  old_any <- rowSums(A_old > 0, na.rm = TRUE) > 0
  sum(old_any & wide[[new]] == 0, na.rm = TRUE) / sum(old_any)
}

data.table(
  recent_map = c("GAEZ+2015", "LUH2", "MIRCA-OS"),
  sees_zero_where_older_detects_pct = round(100 * sapply(recent_t, pooled_blind)))

##   recent_map sees_zero_where_older_detects_pct

```

##	<char>	<num>
## 1:	GAEZ+2015	19
## 2:	LUH2	83
## 3:	MIRCA-OS	17

3 Reviewer 2

3.1 R2.1: census products agree because they share source data, not reality

The five FAO-statistics products share definitions and much of their source data. We compare their existential disagreement against all five-map subsets.

```
census_ev <- exist_union(wide, census)

combs5 <- combn(maps, 5, simplify = FALSE)
ev5 <- sapply(combs5, function(cc) exist_union(wide, cc))

data.table(
  quantity = c("Census-5 (GMIA, MIRCA2000, MIRCA-OS, SPAM, GAEZ+2015)",
    "All 252 five-map subsets: minimum",
    "All 252 five-map subsets: median",
    "All 252 five-map subsets: maximum"),
  existential_pct = round(100 * c(census_ev, min(ev5), median(ev5), max(ev5))),
  note = c(sprintf("percentile %.0f%%", 100 * mean(ev5 <= census_ev)),
    "", "", ""))
```

```
##                                quantity existential_pct
##                                <char>                <num>
## 1: Census-5 (GMIA, MIRCA2000, MIRCA-OS, SPAM, GAEZ+2015)      24
## 2:                      All 252 five-map subsets: minimum      24
## 3:                      All 252 five-map subsets: median      80
## 4:                      All 252 five-map subsets: maximum      90
##          note
##          <char>
## 1: percentile 0%
## 2:
## 3:
## 4:
```

3.2 R2.2: remote-sensing and census products are not a subset of one another

If remote sensing simply mapped a superset of census irrigation, it would never record zero where a census product detects irrigation. We test both directions and the disagreement within each family.

```
any_pos <- function(cols) rowSums(as.matrix(wide[, ..cols]) > 0, na.rm = TRUE) > 0
cP <- any_pos(census)
rP <- any_pos(rs)

data.table(
  quantity = c("RS records zero where census detects irrigation (% of census-irrigated cells)",
    "Census records zero where RS detects irrigation (% of RS-irrigated cells)",
    "Within RS family: GIAM vs GRIPC existential disagreement (%)",
    "Within census family: five FAO products existential disagreement (%)",
    "Mean global total, census maps (Mha)",
    "Mean global total, RS maps (Mha)"),
```

```

value = c(round(100 * sum(cP & !rP) / sum(cP)),
           round(100 * sum(rP & !cP) / sum(rP)),
           round(100 * exist_union(wide, rs)),
           round(100 * exist_union(wide, census)),
           round(mean(sapply(census, function(m) sum(wide[[m]])))),
           round(mean(sapply(rs, function(m) sum(wide[[m]]))))))

```

```

##                                     quantity
##                                     <char>
## 1: RS records zero where census detects irrigation (% of census-irrigated cells)
## 2:      Census records zero where RS detects irrigation (% of RS-irrigated cells)
## 3:                                     Within RS family: GIAM vs GRIPC existential disagreement (%)
## 4:                                     Within census family: five FAO products existential disagreement (%)
## 5:                                     Mean global total, census maps (Mha)
## 6:                                     Mean global total, RS maps (Mha)
##      value
##      <num>
## 1:      36
## 2:      17
## 3:      57
## 4:      24
## 5:     244
## 6:     321

```

*# Per-map global totals underlying the census/RS means (which maps each group
uses follows the reviewer's own classification in his comment) -----*

```

data.table(
  group = c(rep("census", length(census)), rep("remote sensing", length(rs))),
  map = c(census, rs),
  total_mha = round(sapply(c(census, rs), function(m) sum(wide[[m]])), 1))

```

```

##           group      map total_mha
##           <char>   <char>   <num>
## 1:      census    gmia     245.8
## 2:      census mirca2000    210.5
## 3:      census  mirca_os    265.5
## 4:      census    spam     219.6
## 5:      census  gaez_v4     280.3
## 6: remote sensing    giam     335.2
## 7: remote sensing  gripc     306.0

```

3.3 R2.3: census products are compared on a common physical extent

The reviewer notes that the census products map different quantities (area equipped, harvested area, monthly growing area). We do not compare these raw quantities: in `code_original_datasets` we reduce every product to its actual physical irrigated extent, so these magnitude differences do not enter the comparison. The variable used for each product is recorded below.


```

provenance <- data.table(
  product = c("GMIA", "SPAM2010", "MIRCA2000", "MIRCA-OS", "GAEZ+2015", "GIAM"),
  variable_used = c("Actually irrigated area (AEI x AAI fraction), not AEI",
    "Physical irrigated area (SPAM 'A'), capped at cell area",
    "Maximum monthly irrigated area (physical extent), not summed",
    "Maximum monthly growing area (physical extent), not summed",
    "Harvested irrigated area, capped at physical cell area",
    "Net actually irrigated area (TAAI), not annualized (AIA)",
  source = c("code_original_datasets.Rmd L254", "L439, L452", "L341", "L643",
    "L382, L399", "L205"))

knitr::kable(provenance, caption = "Variable extracted for each product (code_original_datasets.Rmd). Every product is reduced to its actual physical irrigated extent, so differences between area equipped, harvested area and monthly growing area do not enter the existence comparison.")

```

Table 2: Variable extracted for each product (code_original_datasets.Rmd). Every product is reduced to its actual physical irrigated extent, so differences between area equipped, harvested area and monthly growing area do not enter the existence comparison.

product	variable_used	source
GMIA	Actually irrigated area (AEI x AAI fraction), not AEI	code_original_datasets.Rmd L254
SPAM2010	Physical irrigated area (SPAM 'A'), capped at cell area	L439, L452
MIRCA2000	Maximum monthly irrigated area (physical extent), not summed	L341
MIRCA-OS	Maximum monthly growing area (physical extent), not summed	L643
GAEZ+2015	Harvested irrigated area, capped at physical cell area	L382, L399
GIAM	Net actually irrigated area (TAAI), not annualized (AIA)	L205

4 Reviewer 3

4.1 R3.14: disagreement does not track irrigation practice

The reviewer asks whether irrigation technique (e.g. drip in Spain) drives the disagreement. We report existential and extreme disagreement per country.

```
# Global reference -----
glob <- data.table(country = "GLOBAL", cells = nrow(wide),
  existential_pct = round(100 * exist_union(wide, maps)),
  extreme_pct = round(100 * extreme_frac(wide, maps)))

# By country -----
ctys <- c("India", "Portugal", "Italy", "Pakistan", "China", "Spain",
  "Egypt", "France", "Mexico", "Iran", "United States")

res <- rbindlist(lapply(ctys, function(cn) {
  s <- wide[country == cn]
  if (nrow(s) == 0) return(NULL)
  data.table(country = cn, cells = nrow(s),
    existential_pct = round(100 * exist_union(s, maps)),
    extreme_pct = round(100 * extreme_frac(s, maps)))
}))

res <- rbind(glob, res[order(-extreme_pct)])

knitr::kable(res, caption = "Existential and extreme disagreement by country (0.2 deg). Existential disagreement is the percentage of countries that disagree with the global average on the existence of a given species. Extreme disagreement is the percentage of countries that disagree with the global average on the extreme of a given species' range.")
```

Table 3: Existential and extreme disagreement by country (0.2 deg). Existential: of the cells where at least one map detects irrigation, the percentage in which at least one map detects none. Extreme: percentage of cropland cells whose $\tau_{\max}$ exceeds ten times the detectability threshold. Spain (the highest-drip country) is not a maximum.

country	cells	existential_pct	extreme_pct
GLOBAL	129628	90	23
India	6517	67	54
Portugal	236	88	49
Italy	819	76	42
Pakistan	1096	62	41
China	12317	71	40
Spain	1336	84	31
Egypt	304	69	31
France	1605	95	29
Iran	2931	88	29
Mexico	3126	92	25
United States	17087	93	15

5 Session information

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.3.1
##
## Matrix products: default
## BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/Libraries/libBLAS.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
##
## locale:
##  [1] C
##
## time zone: Europe/London
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] magrittr_2.0.5   terra_1.8-93     here_1.0.2      data.table_1.18.4
##
## loaded via a namespace (and not attached):
##  [1] vctrs_0.7.3      cli_3.6.6        knitr_1.51       rlang_1.2.0
##  [5] xfun_0.55        otel_0.2.0       generics_0.1.4   S7_0.2.2
##  [9] glue_1.8.1       rprojroot_2.1.1  htmltools_0.5.9  scales_1.4.0
## [13] sensobol_1.2.0   rmarkdown_2.30   grid_4.5.2       tibble_3.3.1
## [17] evaluate_1.0.5   fastmap_1.2.0    yaml_2.3.12      lifecycle_1.0.5
## [21] compiler_4.5.2   codetools_0.2-20 dplyr_1.1.4      RColorBrewer_1.1-3
## [25] Rcpp_1.1.1-1.1   pkgconfig_2.0.3  farver_2.1.2     digest_0.6.39
## [29] R6_2.6.1         tidyselect_1.2.1 Rdpack_2.6.6     pillar_1.11.1
## [33] rbibutils_2.4.1  tools_4.5.2      gtable_0.3.6     ggplot2_4.0.3.9000
```